PROJECT PORTFOLIO · 2026

Sujal G Kulkarni

B.Tech Mechanical Engineering · Vellore Institute of Technology, Chennai

Chennai, India | sujalkulkarni028@gmail.com | +91 91086 76376
GitHub: github.com/sujalkulkarni028-cpu

01 AR-Integrated Assembly Path, Simulation and Execution Method for Digital Twin-Based Component Study

Browser-based AR lab with bare-hand gesture control for CAD assembly, disassembly and training · Capstone

02 Natural Fiber Composites (Funded Project)

Palm and banana fiber reinforced polyester laminates with nano-filler, fiber treatment and characterisation

03 Steady-State Thermal Analysis of Advanced Composite Cooling Fins

Ansys comparison of five fin materials for peak temperature and thermal gradient

04 Nitinol-Actuated Robotic Hand

3D-printed tendon-driven hand actuated by shape memory alloy wires with Arduino PWM heating

05 Design, CNC/CAM Manufacture and Factory Simulation of a Monoblock Water Pump

Computer Integrated Manufacturing project: Fusion CAD/CAM, six-setup CNC machining and a Simio mass-production factory

CAD & Simulation	SolidWorks, Autodesk Fusion (CAD/CAM), Ansys Mechanical & APDL, DWSIM, Simio
Programming	Python (NumPy, Pandas), JavaScript (Three.js), MATLAB, Arduino
Lab & Fabrication	Composite hand lay-up, FDM 3D printing, SMA actuation, welding and casting

01 · CAPSTONE PROJECT · AR / HCI / DIGITAL TWIN

AR-Integrated Assembly Path, Simulation and Execution Method for Digital Twin-Based Component Study

Team: Geevargese John, Sujal G Kulkarni, A. Harshvardhan · **Guide:** Eswaran M, School of Mechanical Engineering, VIT Chennai · 2026 · **Code:** github.com/sujalkulkarni028-cpu/holohand-engineering-lab

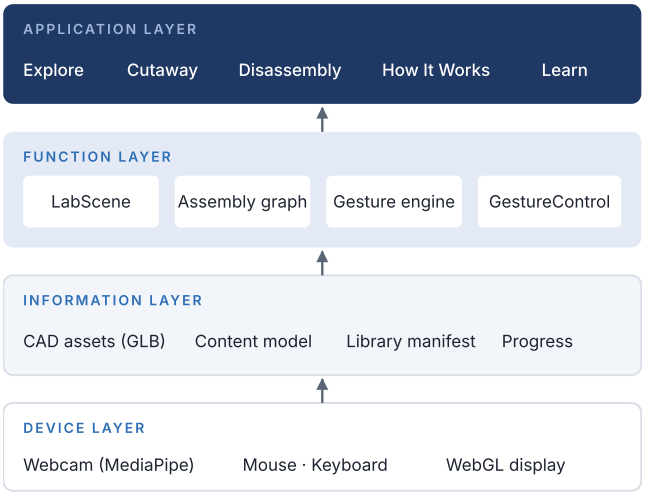
PROBLEM

Industrial AR assembly systems (e.g. HoloLens 2 + Unity) reduce errors and cognitive load, but cost thousands of dollars per seat and are programmed for a single product. Students still learn machine assembly from drawings and the occasional cut-away model.

WHAT WE BUILT

HoloHand Engineering Lab, a 3D lab that runs in the browser. CAD assemblies are stored as digital-twin models and can be inspected, cut away, exploded, disassembled and reassembled with bare-hand gestures from an ordinary webcam.

- Any GLB/glTF assembly from CAD becomes an assembly graph with per-part home transforms and automatically computed explode vectors.
- MediaPipe 21-point hand tracking feeds a rule-based gesture engine with hysteresis and multi-frame stability filtering.
- Each part is linked to an information model covering its function, location, connections and role in the cycle, plus guided lessons, an animated operating cycle and quizzes.



Four-layer architecture, following the information/function/application framework of Zhang et al. (2025). Mouse, keyboard and gestures all call the same functions.

22 parts
two-spool turbofan, 145,156 triangles; plus a 9 : 1 two-stage gearbox

67–267 ms
gesture stabilisation at 30 fps

4.6 µs
gesture classification per frame; 0.18 ms ray-cast part pick

~120×
lower cost per seat than a HoloLens 2 setup; 100% open source

GESTURE → ACTION MAPPING

Gesture	Action	Rule
Open hand, moving	Rotate assembly	ignored below 1.5 px
Point	Highlight / select part	0.8 s dwell
Pinch on part	Grab and pull out	engage $\rho<0.32$, release $\rho>0.50$
Fist, hold	Reassemble	0.7 s hold
Two hands	Zoom	$z = z_o(D/D_o)^s$
Peace sign, hold	Next lesson step	0.5 s, 1.6 s cool-down

ENGINEERING HIGHLIGHTS

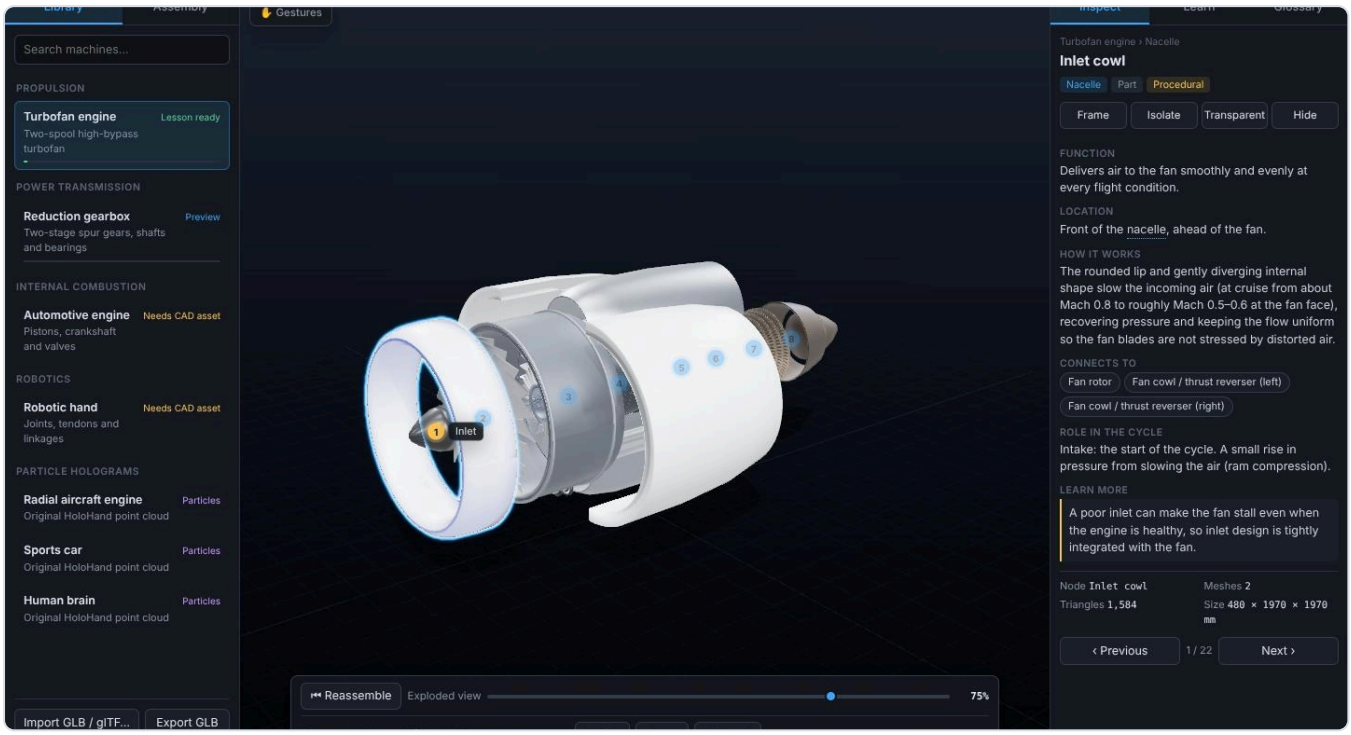
- **Explode model** for any CAD hierarchy: $x_i = x_{i,home} + e \cdot v_i + \delta_i$, with vectors scaled by tree depth.
- **Gesture features** based on ratios, so they work at any hand size or distance from the camera, with a dead band between "extended" ($r>1.50$) and "curled" ($r<1.32$).
- **Renderer** rebuilt from a point cloud into a solid assembly graph, so parts can be picked exactly and every view change can be undone.
- **Gearbox kinematics** from tooth counts: $i = (42/14)(48/16) = 9$.

JavaScript · Three.js / WebGL · MediaPipe Hand Landmarker · glTF / GLB · Node.js · Playwright testing · Digital twin · HCI

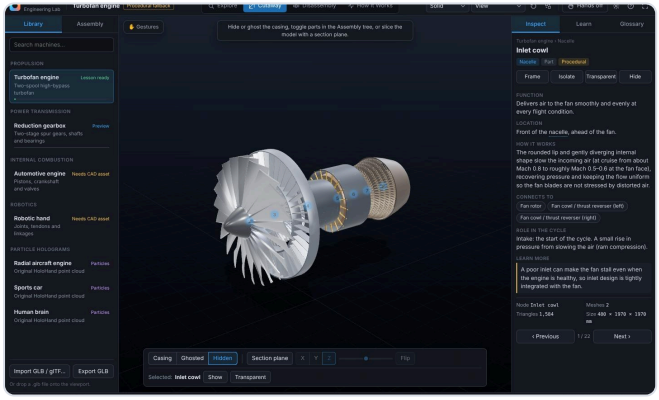
01 · CAPSTONE PROJECT · WORKING MODEL

HoloHand Engineering Lab in Action

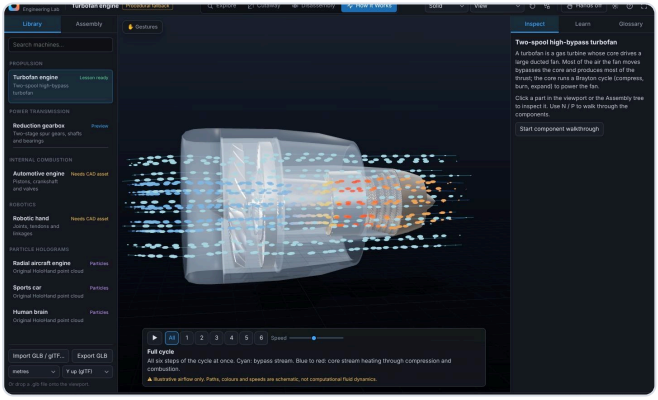
Screenshots from the running application, showing a two-spool turbofan (22 parts) in three of the five modes



Disassembly mode. The turbofan is exploded to 75 % and the inlet cowl is selected (blue outline). The inspector panel explains its function, location, operating principle and connected parts.

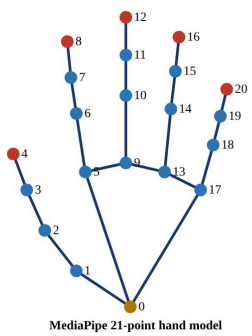


Cutaway mode. The casing is hidden to show the compressor, combustor and turbines.



How It Works mode. Bypass (cyan) and core (blue to red) airflow through the engine.

HAND GESTURE CONTROL

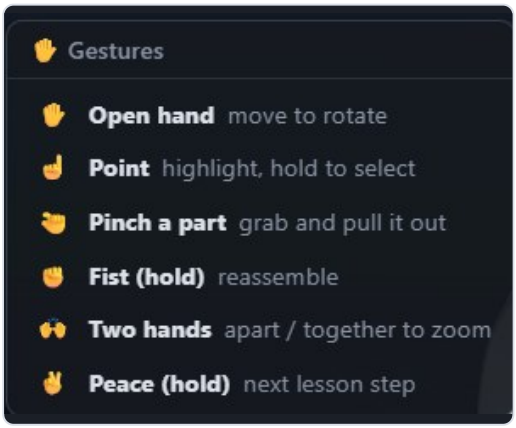


MediaPipe 21-point hand model

Gesture	Rule on finger ratios r_k (index, middle, ring, pinky)
OPEN	all four $r_k > 1.50$
POINT	index > 1.50; middle, ring, pinky < 1.32
PEACE	index, middle > 1.50; ring, pinky < 1.32
FIST	all four $r_k < 1.32$
PINCH	$\rho = \lvert p4 - p8 \rvert / \lvert p0 - p9 \rvert < 0.32$ (release > 0.50)

$r_k = |\text{tip}_k - \text{wrist}| / |\text{MCP}_k - \text{wrist}|$, landmarks mirrored and aspect-corrected. A gesture becomes stable after 3 consecutive frames (PINCH 2, PEACE 8).

MediaPipe 21-point hand model and the finger-ratio rules ($r_k = |\text{tip} - \text{wrist}| / |\text{MCP} - \text{wrist}|$) used to classify each gesture.



On-screen gesture guide, as the learner sees it in the app.

02 · FUNDED RESEARCH PROJECT · MATERIALS

Natural Fiber Reinforced Polyester Composites

Focus: palm and banana fiber reinforcement, nano-filler addition, chemical fiber treatment and material characterisation



Freshly cast specimens in the mould (strips and squares for property testing)



Fiber-loaded laminate specimens after partial cure



Banana fiber / polyester batch cast by hand lay-up

OBJECTIVE

Develop low-cost, sustainable composites from agricultural fiber waste (palm and banana) in a polyester matrix, and study how fiber content, chemical treatment and nano-filler addition change the material's properties.

PROCESS

- **Fiber treatment:** untreated fibers compared with alkali (NaOH, KOH), acid (HCl) and silane treatments to improve fiber–matrix bonding.
- **Formulation:** each 280 g batch weighed by mass fraction of fiber and filler, with catalyst and accelerator added to the polyester resin.
- **Fabrication:** hand lay-up into an open mould with cavities for test specimens, cured at room temperature.
- **Characterisation:** SEM (fracture surface and fiber bonding), FTIR (chemical groups after treatment), TGA (thermal stability) and XRD (crystallinity).

SAMPLE FORMULATIONS (280 G BATCH)

Sample	Fiber	Nano-filler	Polyester
S-2	Banana 4 % (11.2 g)	None	263.2 g
S-8	Palm 4 % (11.2 g)	Nano hBN 1 % (2.8 g)	257.6 g
S-9	Palm 2 % (5.6 g)	None	266.0 g

Catalyst and accelerator: 2.8 g each (1 %) for S-2; 4.2 g each (1.5 %) for S-8 and S-9.

TAKEAWAYS

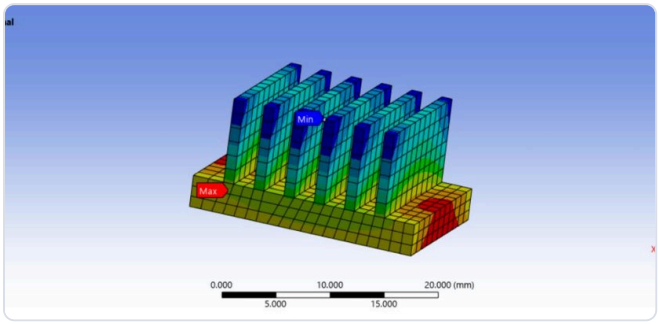
- Designed a series of formulations so that fiber type, fiber fraction and filler content could each be varied one at a time.
- Learned practical lay-up control: resin-to-catalyst ratio, trapped air bubbles and spread across the mould all visibly affect specimen quality.

Natural fibers · Unsaturated polyester · Nano hBN · Hand lay-up · SEM · FTIR · TGA · XRD · Sustainable materials

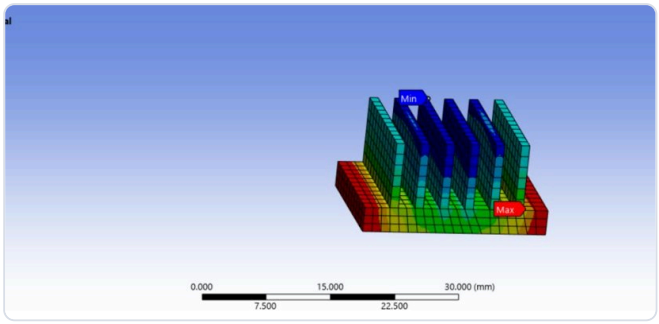
03 · HEAT AND MASS TRANSFER (BMEE402L) · SIMULATION

Steady-State Thermal Analysis of Advanced Composite Cooling Fins

Team: Mirza Fadil Baig, Satyajeet Puthuppully Ajayan, Sujal Kulkarni · Winter 2025–26 · **Tool:** Ansys 2025 R2



Temperature contour on the finned heat sink



Heat-flux distribution across the fin array

OBJECTIVE

Compare the steady-state performance of five fin materials (carbon nanotubes, diamond-reinforced copper, copper–graphite composite, graphite–aluminium co-extrusion and standard aluminium) on the same 22 × 11 × 15 mm fin geometry, ranking them by peak temperature and internal temperature gradient (ΔT). Heat-flux outputs were converted to the same units (W/mm²) so the runs could be compared directly.

≈ 2.7 °C
lower peak temperature with CNT and diamond–Cu than with aluminium

0.006 °C
internal ΔT for CNT, against 0.104 °C for aluminium

~38 %
lower ΔT with copper–graphite than with aluminium

5
materials ranked on one fixed geometry

RESULTS

Rank	Material	Max temp (°C)	Min temp (°C)	ΔT (°C)	Max heat flux (W/mm²)
1	Carbon nanotubes	30.638	30.632	0.006	2.699×10^{-3}
2	Diamond-reinforced copper	30.654	30.621	0.033	2.695×10^{-3}
3	Copper–graphite composite	33.304	33.240	0.064	2.694×10^{-3}
4	Graphite–aluminium	33.329	33.235	0.094	2.918×10^{-3}
5	Standard aluminium (baseline)	33.331	33.227	0.104	2.692×10^{-3}

RECOMMENDATIONS

Maximum heat extraction
Carbon nanotubes, with almost no internal thermal resistance. Diamond-reinforced copper performs almost identically if CNTs can't be manufactured.

Cost vs performance
Copper–graphite composite: a clear improvement over aluminium in heat spreading, without exotic processing.

Weight-sensitive (aerospace)
Graphite–aluminium co-extrusion spreads heat better than aluminium while staying light.

Ansys Steady-State Thermal · Conduction & convection · Material selection · Thermal management

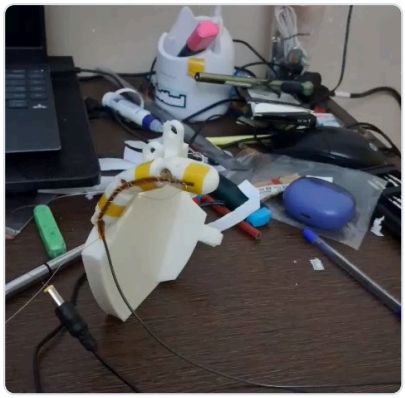
04 · DESIGN & PROTOTYPING · MECHATRONICS

Design and Implementation of a Nitinol-Actuated Robotic Hand

Team: Dhruv Swarup, Sujal G K, Viswajit G, Aryan, Fadil · **Tools:** SolidWorks, FDM 3D printing (PLA), Arduino



SolidWorks model of the anthropomorphic hand



Printed PLA prototype with SMA tendons



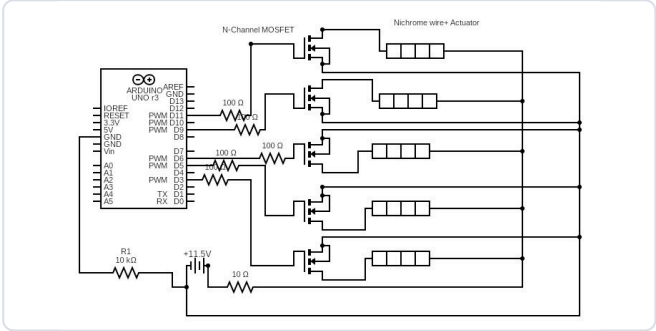
Nitinol wire wrapped in Kapton tape and wound with nichrome

CONCEPT

Replace motors and pneumatics with shape memory alloy (Nitinol) wire. When heated, the wire contracts and pulls tendons that flex the finger joints, giving silent actuation with a high power-to-weight ratio in a compact, low-cost hand.

DESIGN & BUILD

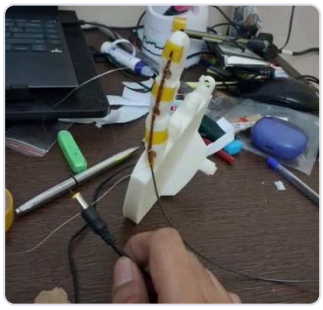
- Modular hand designed in SolidWorks and printed in PLA. Grooves route two wires through each finger, one to bend it and one to straighten it.
- Each Nitinol wire is insulated with Kapton tape and wound with nichrome wire (0.7 cm per wire, 12 V, 1 A supply) for direct heating.
- An Arduino drives the heaters with PWM through N-channel MOSFETs, one channel per actuator.



Control schematic: Arduino PWM outputs switching nichrome heaters through MOSFETs



Single-finger test: heater on



Finger returning to the extended position

RESULTS & LESSONS

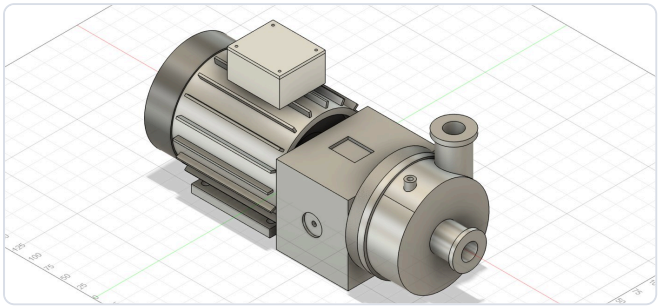
- The single-finger test confirmed tendon-driven flexion and extension by heating the SMA.
- Response speed was limited by heat-up and cool-down time. It can be improved by preheating, thinner insulation and thinner Nitinol wire.
- The wire did not fully recover its strain because the design pushed it beyond the 8–10 % recoverable strain range, which is a key constraint for the next design.
- **Next steps:** tactile sensing, closed-loop temperature control and carbon-fiber-reinforced PLA for a stiffer structure.

SolidWorks · Shape memory alloy · Tendon-driven mechanism · Arduino PWM · MOSFET drive · PLA / FDM

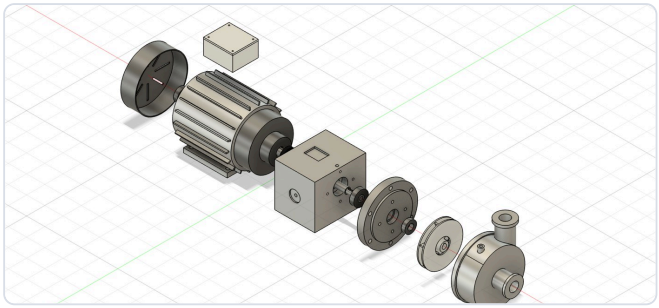
05 · COMPUTER INTEGRATED MANUFACTURING · THEORY + LAB PROJECT

Design, CNC/CAM Manufacture and Factory Simulation of a 0.5 HP Monoblock Water Pump (MWP-370)

Tools: Autodesk Fusion (CAD, CAM, Python API scripts), Simio (discrete-event simulation) · VIT Chennai · October 2026



MWP-370 pump assembly in Fusion (15 components)



Exploded view in the order the parts go together on the assembly line

20 m · 1,800 L/h
design head and flow, checked by hand calculations

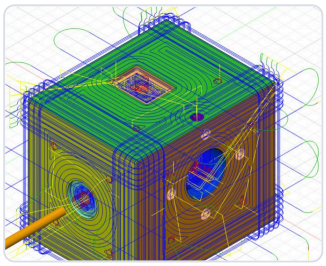
31 ops, 0 clashes
six-setup CAM of the 6-face block, verified in simulation

682 pumps
per 80 h in the Simio factory (≈ 40,920 a year)

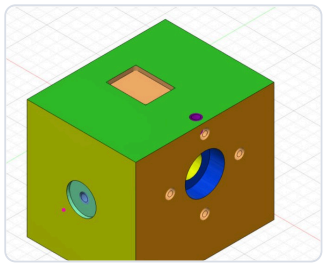
₹3,779
unit cost; ₹4.42 crore plant, 2.6-year payback

PRODUCT DESIGN

A 0.37 kW, 2,880 rpm pump for lifting water from a sump to an overhead tank. Hydraulic power, impeller tip speed, shaft size and bearing life were calculated by hand and then modelled in Fusion as 7 made parts, 1 sheet-metal cowl and 7 bought-out items.



Toolpaths on all six faces



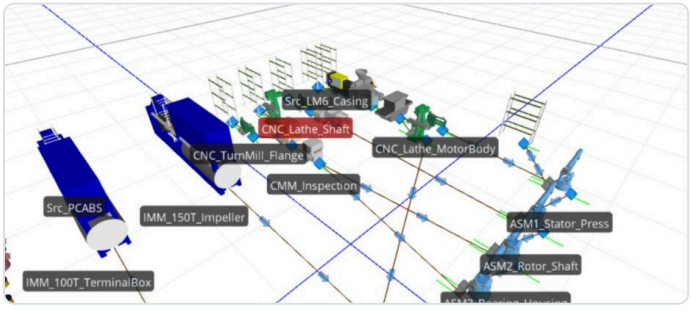
Simulated finished block

CNC MACHINING & CAM

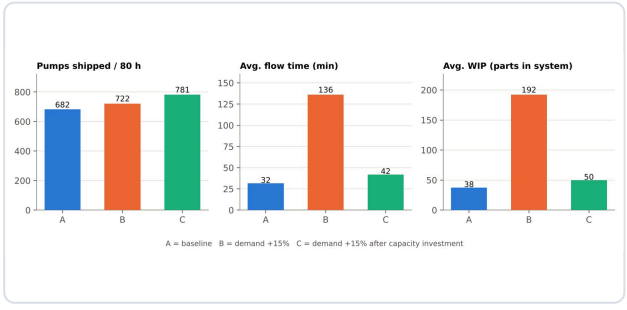
- **Three CNC parts:** an SS410 shaft (turning), a cast-iron motor flange (turning and drilling) and an aluminium bearing-housing block machined on all six faces.
- The block was programmed as **six setups**, with the stock carried from one setup to the next. Fanuc G-code was posted for every setup.
- The first program gave **268 tool-holder collisions**. Reducing the cut depth and changing the bore to a drilling operation cleared all of them before any metal would be cut.

FACTORY SIMULATION (SIMIO)

- The model covers CNC, die-casting and injection-moulding cells, a 7-station assembly line, testing, rework and packing.
- It showed the 6-face VMC (91 % busy) and the flange turn-mill (85 %) as the **bottlenecks**. With 15 % more demand, flow time rose from 32 to 136 min; adding one VMC and one turn-mill raised output to 781 pumps.
- **Quality:** control plan with SPC; Cp 1.39 / Cpk 1.33 on the Ø40 H7 bearing seat.



3D Simio model of the pump factory: CNC, moulding and die-casting cells feeding the assembly line



Output, flow time and WIP for the three scenarios: baseline, +15 % demand, and +15 % with extra machines

Autodesk Fusion CAD/CAM · Multi-setup CNC milling · G-code post-processing · Simio · Bottleneck analysis · SPC · Cp/Cpk · Cost & ROI · DFMA